【数据结构】链表

Tags:

Time Limit: 1 s Memory Limit: 128 MB
Submission：211 AC：31 Score：98.53

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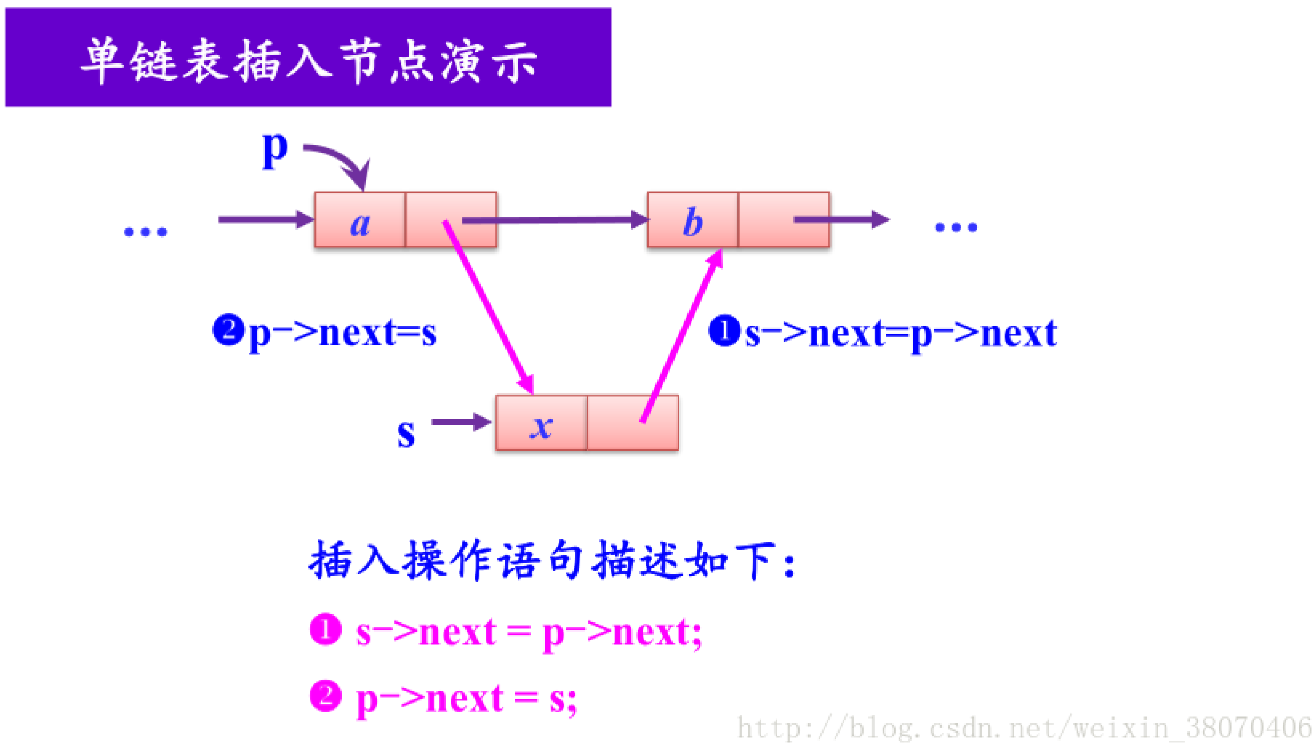
Description

链表的插入流程如下图所示：
第一步：
新插入节点s的next指针指向p节点的next指针

第二步：

将p->next 指向s



完成以上两步之后，即完成了新节点的插入。
节点的删除过程与上图类似。
现在阅读以下程序，并补全

```
Status ListInsert(NHead *head, int pos, ElemType e);
Status ListDelete(NHead *head, int pos);
```

两个函数，使得程序能够完成新节点的插入或删除某个位置的元素

元素下标从0开始

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <math.h>
#define OK 1
#define ERROR 0
#define TRUE 1
#define FALSE 0
typedef int ElemType;
typedef int Status;
typedef struct Node
{
 ElemType data;
 struct Node *next;
}Node;
typedef Node *NPointer;
typedef struct NHead
{
 int length;
 NPointer next;
}NHead;
Status InitList(NHead *head)
{
 head->length = 0;
 head->next = NULL;
 return OK;
}
Status ClearList(NHead *head)
{
 head->length = 0;
 while (head->next)
 {
  NPointer tmp = head->next;
  head->next = head->next->next;
  free(tmp);
 }
 return OK;
}
Status ListIsEmpty(NHead *head)
{
 if (head->length)return FALSE;
 else return TRUE;
}
int ListLength(NHead *head)
{
 return head->length;
}
Status GetElem(NHead *head, int pos, ElemType *e)
{
 NPointer now = head->next;
 int cot = 0;
```

```
 while (now&&cot < pos)
 {
  now = now->next;
  cot++;
 }
 if (!now || cot >= pos)return ERROR;
 *e = now->data;
 return OK;
}
int LocateElem(NHead* head, ElemType e)
{
 int cot = 0;
 NPointer now = head->next;
 while (now)
 {
  if (now->data == e) /* 找到这样的数据元素 */
   break;
  now = now->next;
  cot++;
 }
 return cot;
}
void ListVisit(ElemType e)
{
 printf("%d", e);
}
Status ListTraverse(NHead *head)
{
 NPointer now = head->next;
 int cot = 0;
 while (now)
 {
  if (cot)printf(" ");
  ListVisit(now->data);
  cot++;
  now = now->next;
 }
 printf("\n");
 return OK;
}
Status ListInsert(NHead *head, int pos, ElemType e);
Status ListDelete(NHead *head, int pos);
int main()
{
 NHead head;
 InitList(&head);
 int n;
 scanf("%d", &n);
 char ctl[10];
 for (int i = 0; i < n; i++)
 {
  scanf("%s", ctl);
  if (strcmp(ctl, "add") == 0)
  {
   int x, y;
   scanf("%d%d", &x, &y);
   ListInsert(&head, x, y);
  }
  else if (strcmp(ctl, "find") == 0)
  {
   int x;
   scanf("%d", &x);
   int pos = LocateElem(&head, x);
   if (pos == ListLength(&head))puts("No such element");
   else printf("%d\n", pos);
```

```
		}
		else if (strcmp(ctl, "delete") == 0)
		{
			int x;
			scanf("%d", &x);
			if (ListDelete(&head, x))puts("OK");
			else puts("ERROR");
		}
		else if (strcmp(ctl, "traverse") == 0)
		{
			ListTraverse(&head);
		}
		else if (strcmp(ctl, "clear") == 0)
		{

			ClearList(&head);
		}
	}
	return 0;
}
```

你只需要提交两个函数即可

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